

Long-Term Health Forecasting with Wearable Device Data

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Meet the Team

Weaver Goldman



CS/DS BS/MS 2027

Deep Learning, Software Engineering, Data Visualization

I learned about time series feature engineering.

Ian Schneider



DS BS/MS 2027

Time Series Analysis, Machine Learning, Data Visualization

I learned about feature engineering, data engineering, and polars

Abbas Jivan



CS/DS BS/MS 2027

Reinforcement Learning, Applied AI Application

I learned about time series prediction.

Project Vision

1

Investigate Health Forecasting

- Represent “health” using metrics available through wearable devices (e.g., Apple Watch, Fitbit)
- Compare approaches for predicting future health status based on recent health history

2

Visualize Health Trajectories

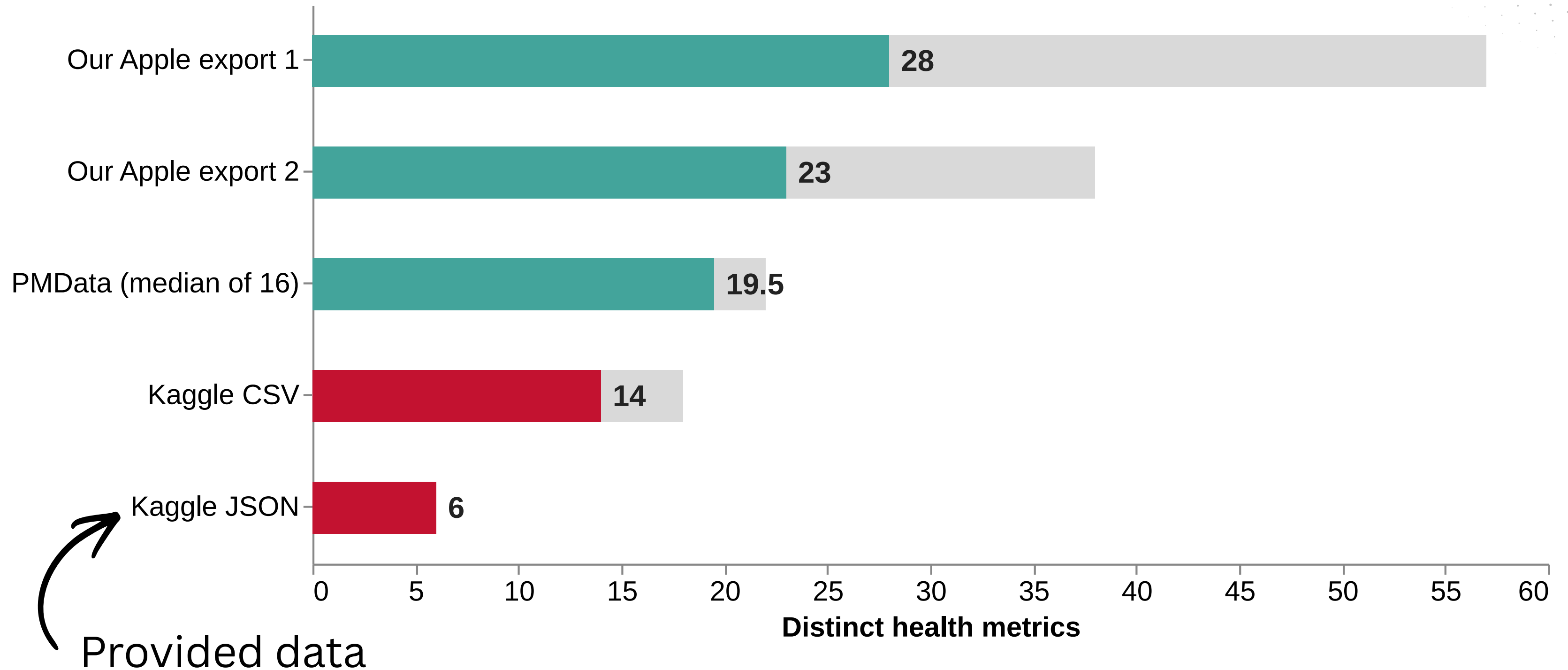
- Develop an interactive dashboard for visualizing a user’s health status
- Enable a user experiment with changes to their routine and how these could impact their future health

Provided Data Overview

Data from Kaggle:

Metric	Unit	Records	Days with data	First day	Last day	Median
HeartRate	count/min	82,474	279	2020-11-17	2025-06-04	76
StepCount	count	15,278	361	2024-06-09	2025-06-04	51
RespiratoryRate	count/min	12,736	264	2024-06-11	2025-06-04	13.5
OxygenSaturation	fraction	5,185	277	2024-06-11	2025-06-04	0.97
BodyMass	kg	282	229	2020-11-15	2025-06-04	96.14
SleepDurationDaily	s	276	276	2024-06-09	2025-06-04	27,792

Supplementing the Provided Data



Data Sources

Kaggle

People	2
Days per person	1,662 and 1,016
Records per person	1.07M and 116K
Metrics on 30+ days	14 and 6
Device	iPhone and wearables

- + Real exports from outside our team
- Few signals, so little to learn from

Personal

People	2
Days per person	3,540 and 1,229
Records per person	2.73M and 2.29M
Metrics on 30+ days	28 and 23
Device	iPhone, watch and scale

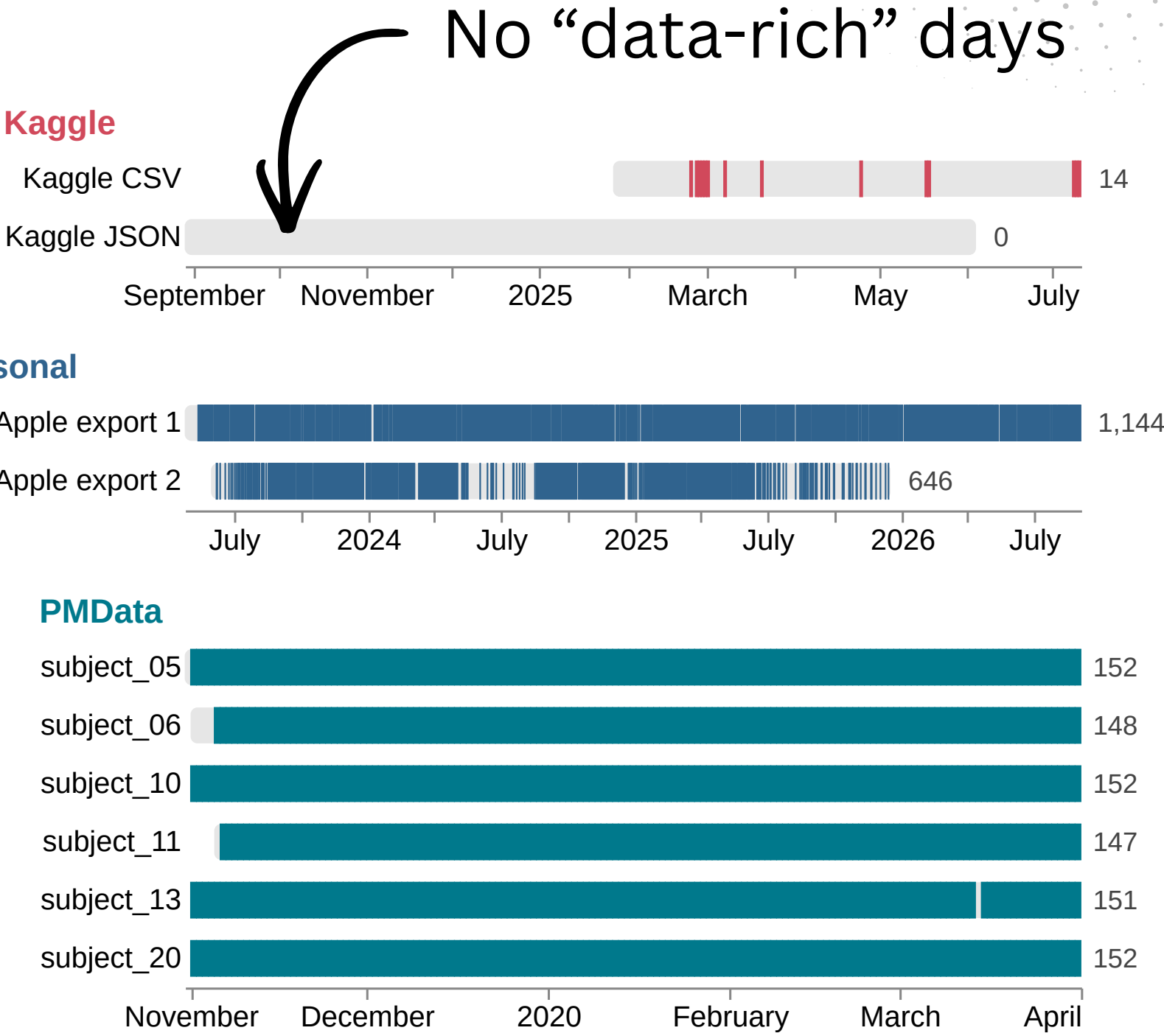
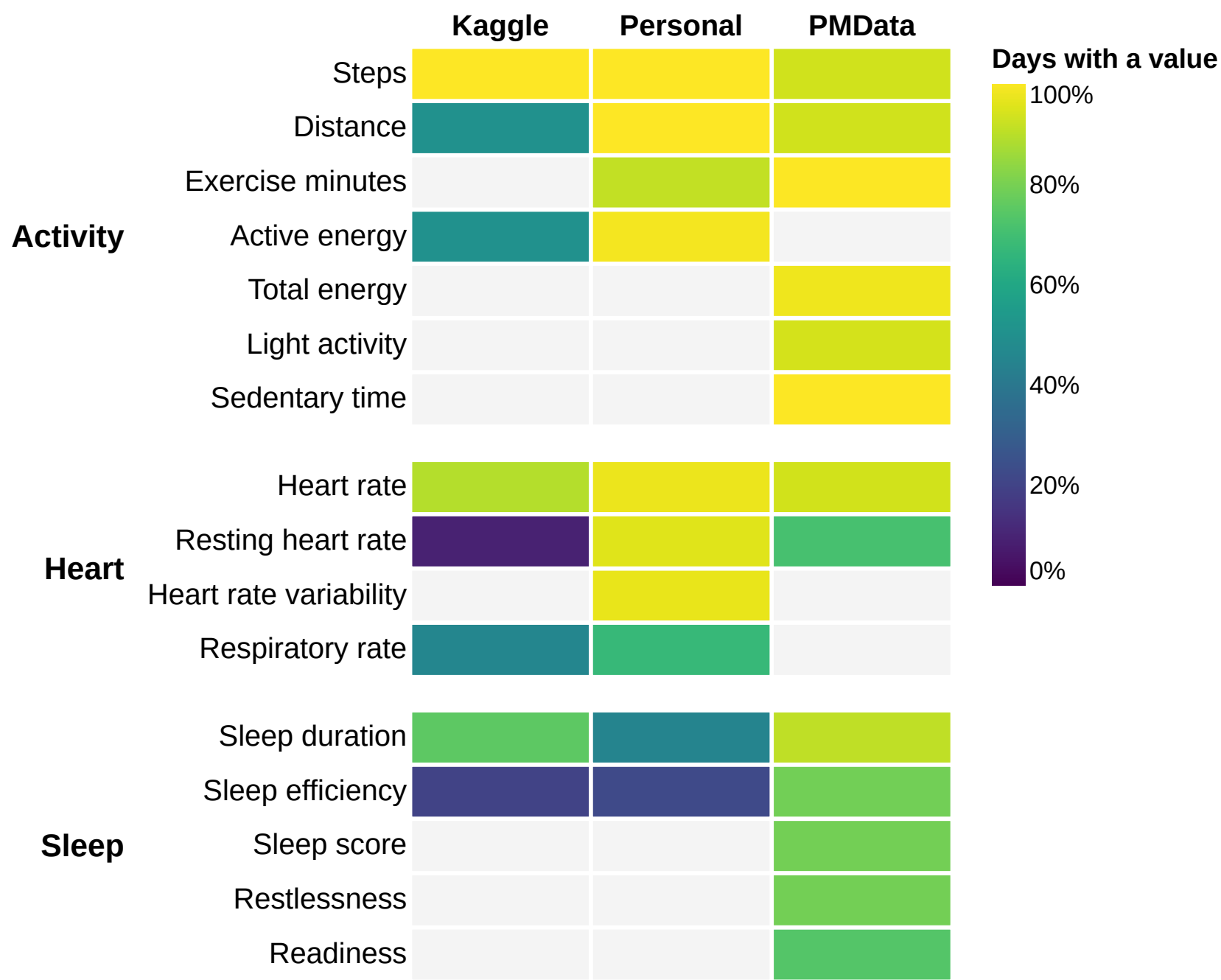
- + The richest and longest records
- Only two people

PMData

People	16
Days per person	139 to 152
Records per person	954K to 2.23M
Metrics on 30+ days	17 to 21
Device	Fitbit and PMSys app

- + Many people on one protocol, plus wellness labels
- Five months only, and one device type

Feature Engineering



Methodology: Embeddings

~16 Health Signals → 2 Dimensions

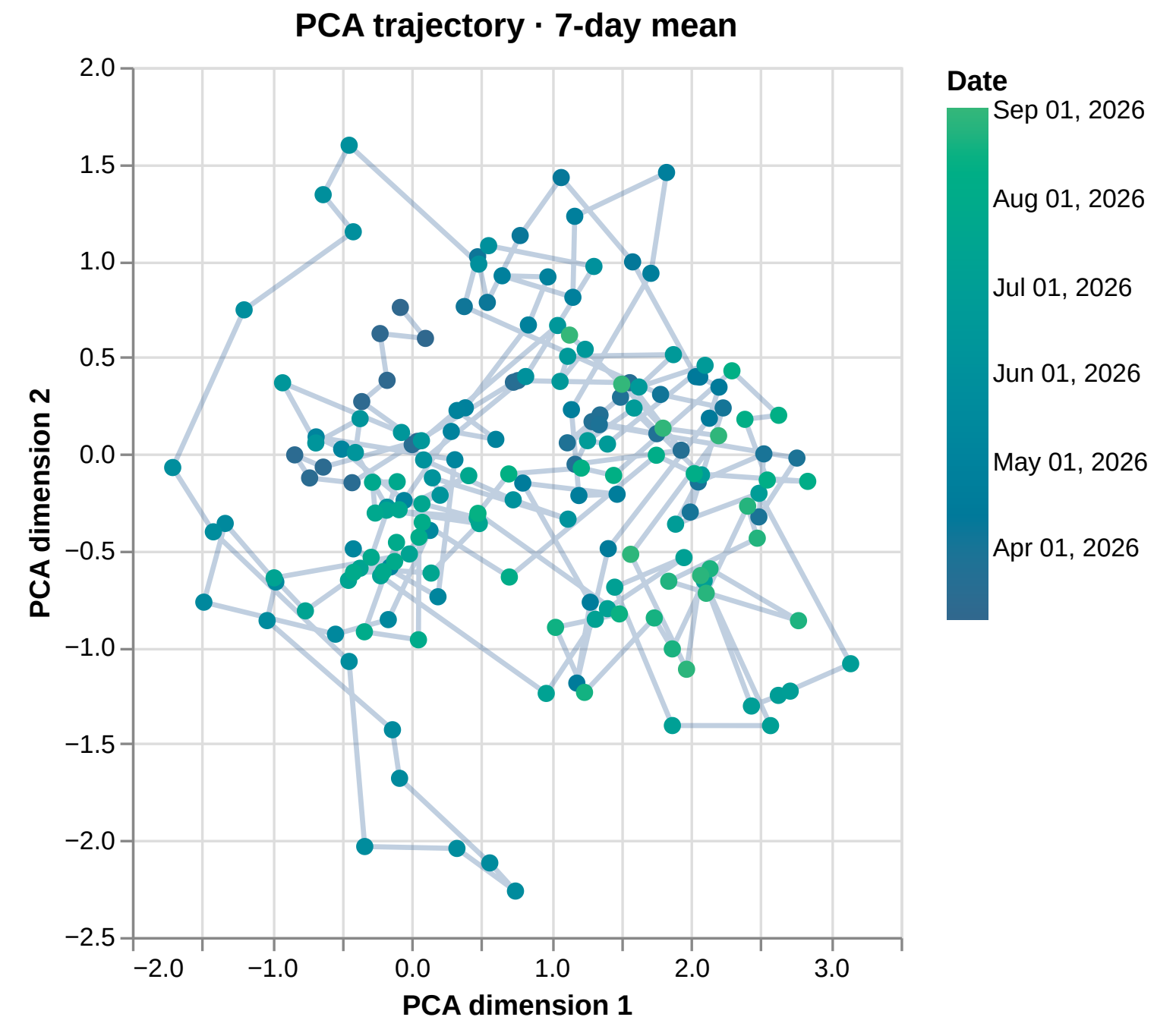
- 62% variance captured
- 0.700 Held-out RSME

Main Takeaway

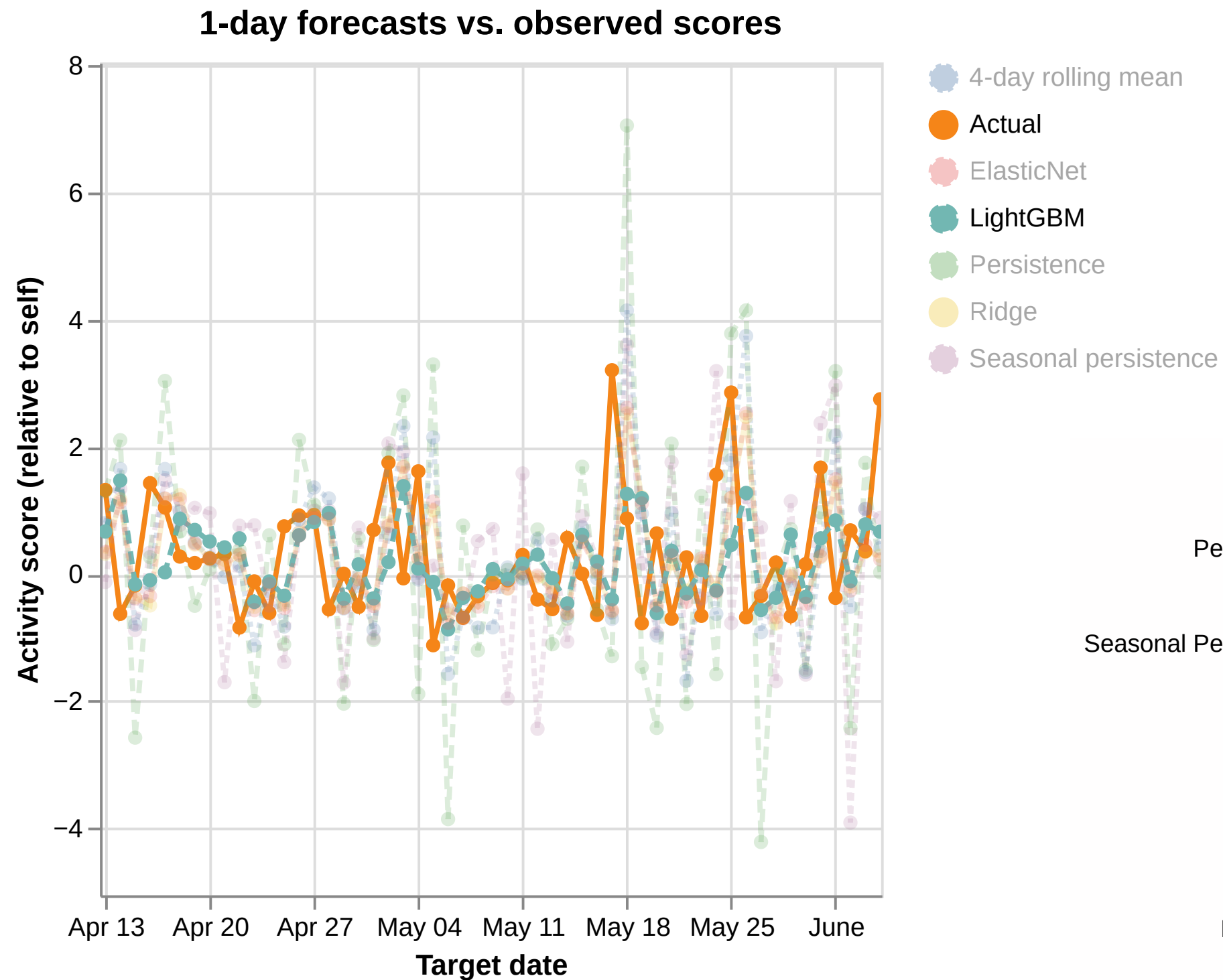
- Embeddings capture changes in a person's combined activity pattern over time

Gaps

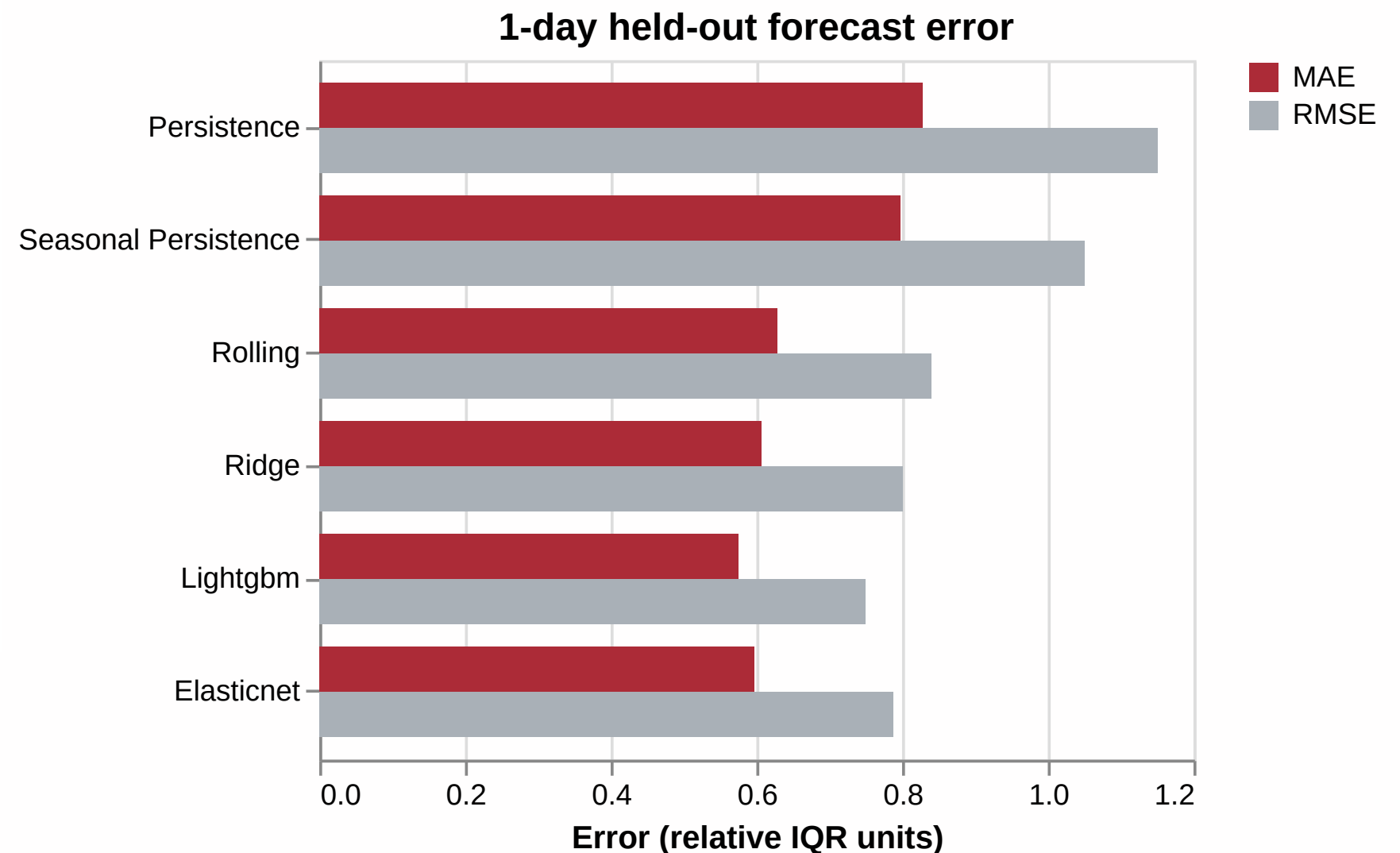
- Amount of data is still insufficient for population level conclusions



Methodology: Forecasting

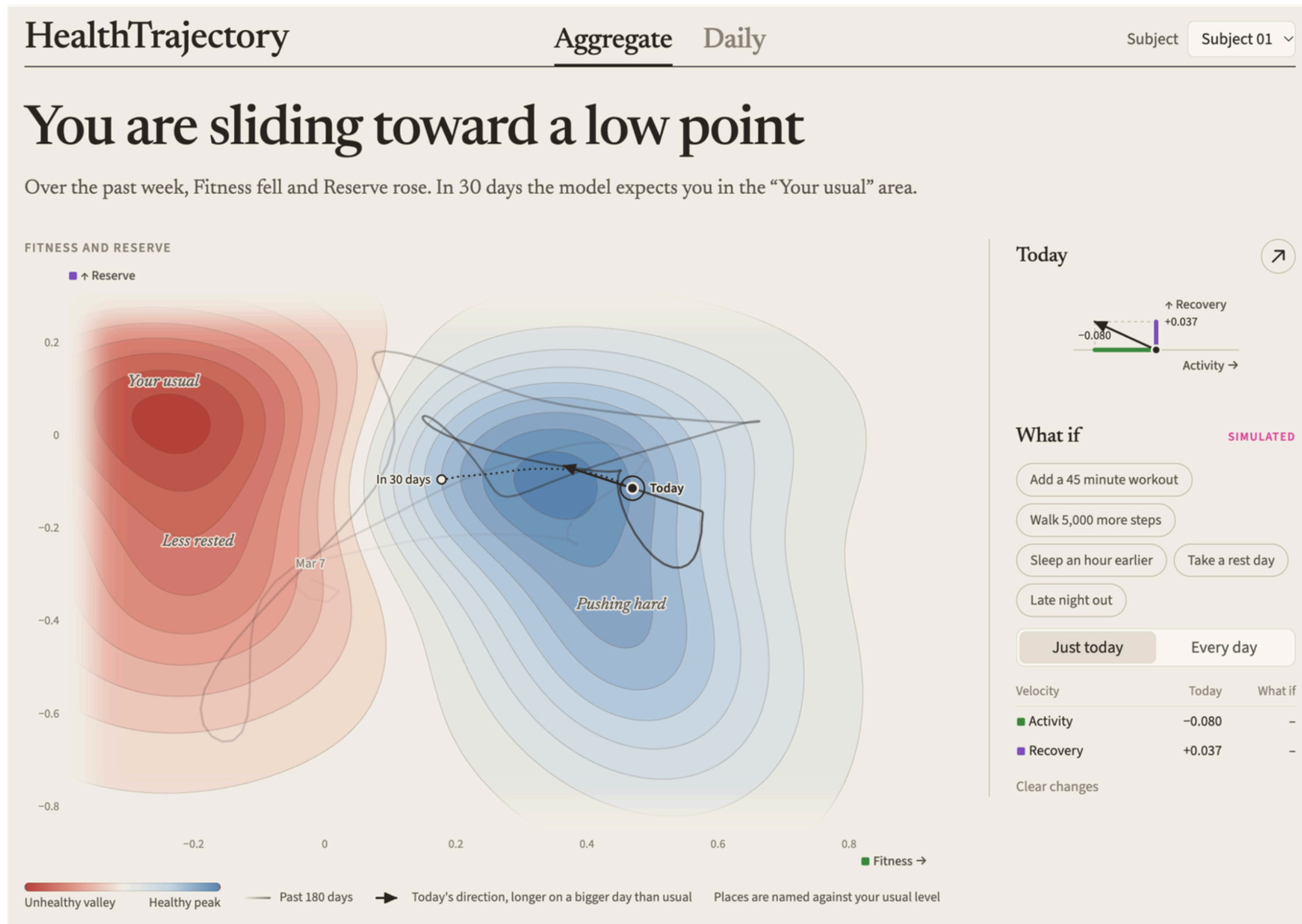


- All models were statistically significantly better than persistence (Predicting the same exact value as the day before)
- No ML model performed statistically significantly better than the other



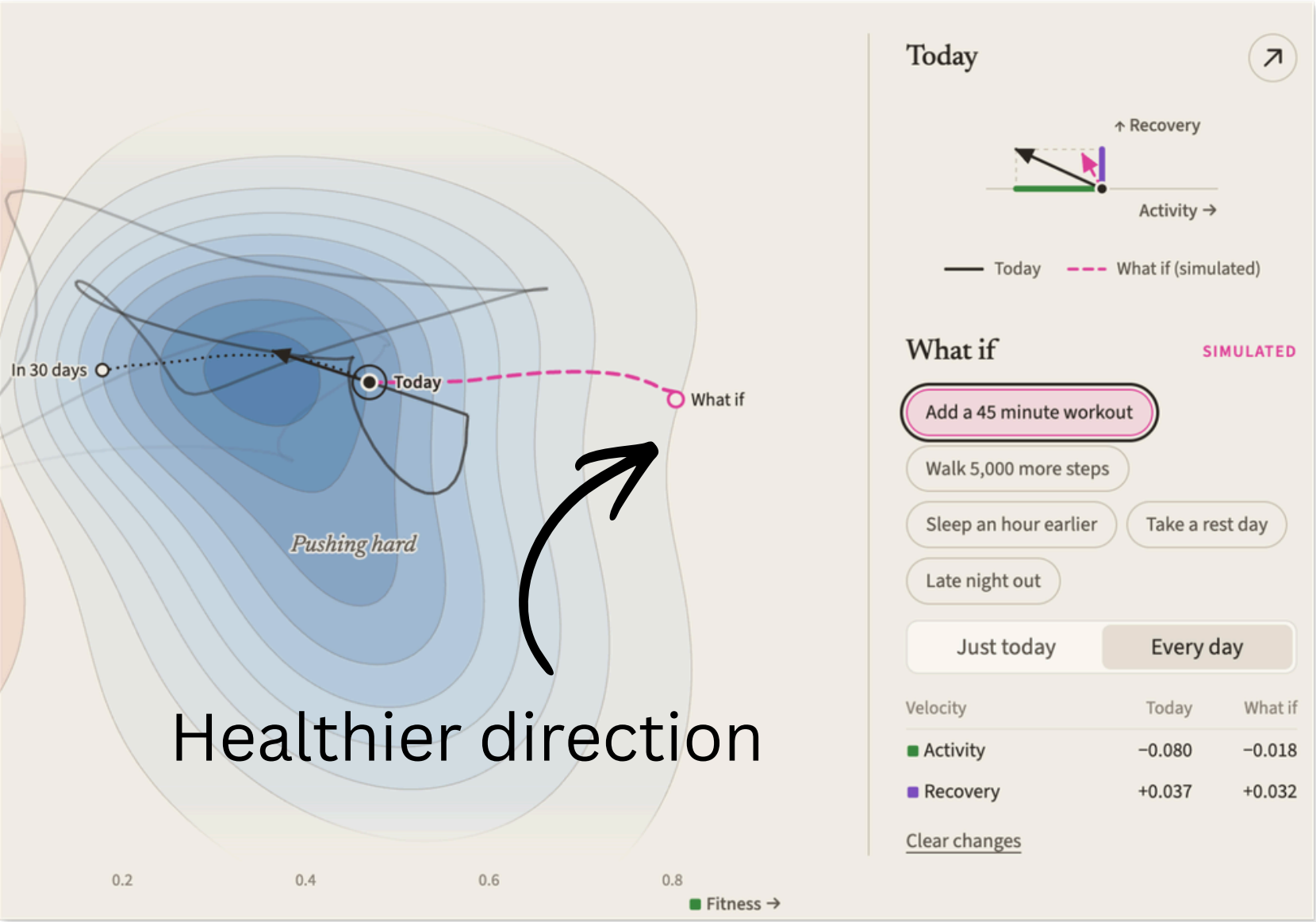
LightGBM had the lowest MAE for 1-day horizon forecasts of activity scores

Demo – Dashboard

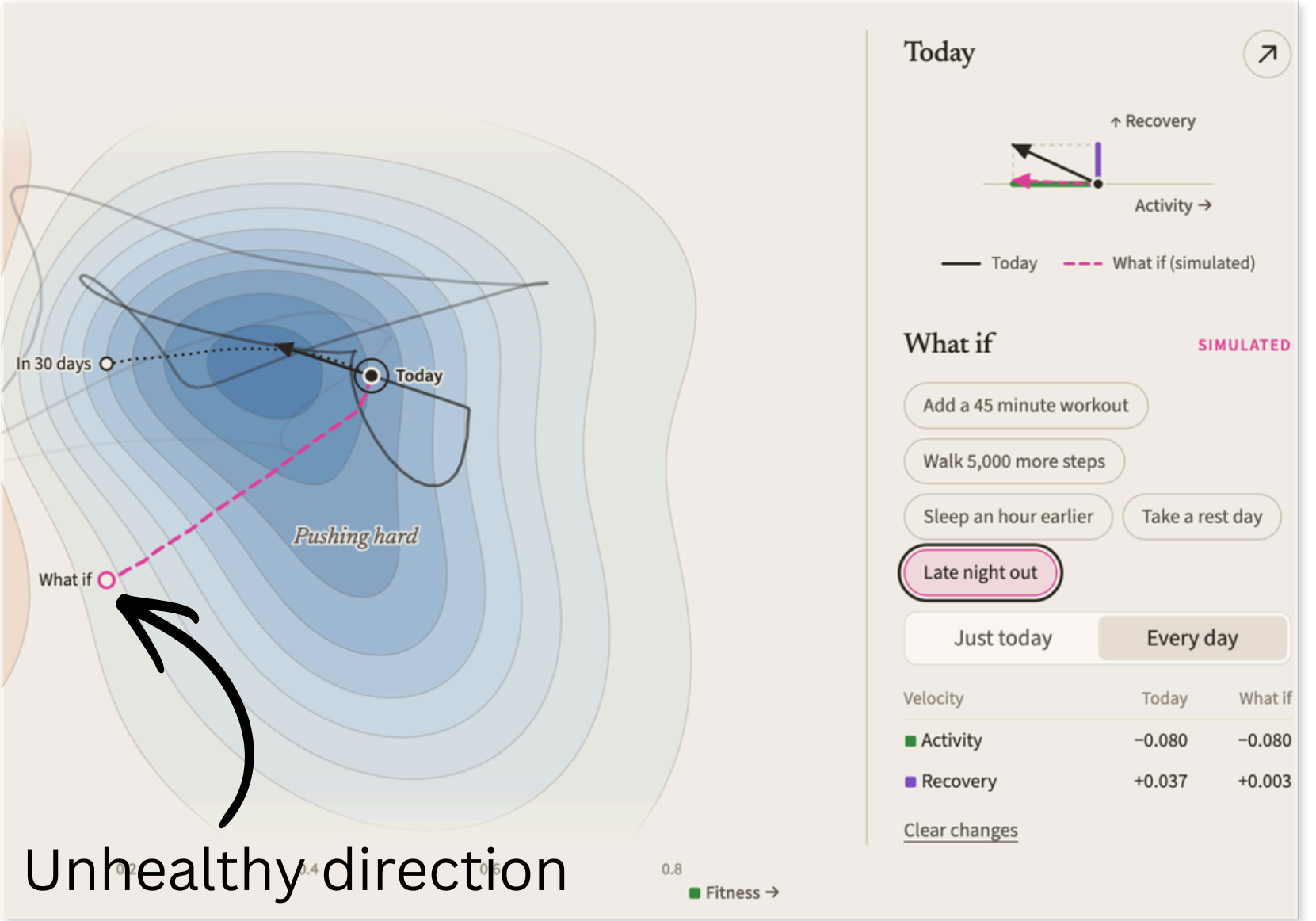


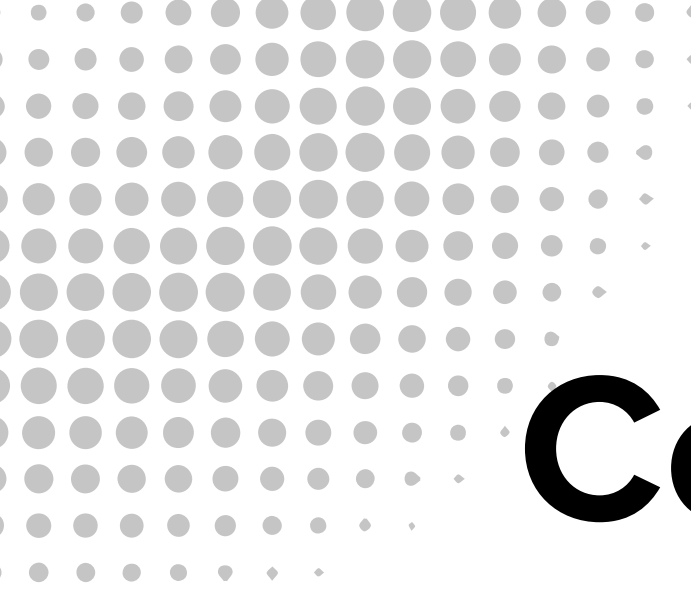
Dashboard – What If?

Add a 45-minute workout
every day



Late night out
every day





Conclusions & Next Steps

Findings

- It is possible to predict future health better than by random chance
- High short-term variation, longer-term patterns
- There appears to be high variation between individuals

Next Steps

- Collect richer and more diverse data
 - Incorporate user embeddings into the prediction process
 - Experiment with larger, more advanced neural-network architectures
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